

Designing bent functions using the Maiorana-McFarland secondary construction

Juan Carlos Ku-Cauich · Javier
Díaz-Vargas

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Abstract We utilise the Maiorana-McFarland secondary construction to generate bent functions. Our process begins with a bent function of dimension $s-r$ and results in a new function of dimension $s+r$. To achieve this, we first consider affine spaces from which we can derive bent functions. Furthermore, we derive an expression to obtain bent functions by applying the case $r = 1$ multiple times. We demonstrate that the two resulting expressions are distinct. In nearly all constructions, the bent functions we obtain are balanced when the domain is restricted to vectors with an even Hamming weight. Finally, we present an example algorithm, focusing on the case where $r = 2$, as well as another algorithm that employs $r = 1$ twice.

Keywords: Bent functions, Maiorana-McFarland, Affine spaces, Balancedness

1 Introduction

Bent functions are a special class of Boolean functions [16] that achieve maximum non-linearity, maximising the distance from affine functions, measured by Hamming distance. These functions are particularly important in cryptography, as they help resist linear attacks. Various methods have been developed to identify good cryptographic properties [1], [14]. Given the large size of the search space (there are 2^{2^n} Boolean functions over the domain $\mathbb{F}_2^n$), there is significant interest in discovering unique characteristics of Boolean functions.

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Juan Carlos Ku-Cauich
Computer Science, CINVESTAV-IPN, Mexico City, Mexico, E-mail: jcku@cs.cinvestav.mx

Javier Díaz-Vargas
Facultad de Matemáticas, UADY, Mérida Yucatán, Mexico, E-mail:
javier.diaz@correo.uady.mx

Distinct constructions can be found in Carlet's book [4] and in Tokareva's book [17, Chapter 8].

We have a unique method for arranging the elements of $\mathbb{F}_2^s$. This approach is significant because it allows us to define and better understand the structure of affine spaces, which in turn helps illustrate various results. Affine spaces are employed in the construction of plateaued, normal, and weakly normal functions. An example of normal functions is the Maiorana-McFarland construction [4], [7], [13], [17]. In [15], a connection between plateaued functions and bent functions is established. They use the relation $f(\bar{x}, \bar{y}) = g(\bar{x})$. We also use the same relation, but we find bent functions f defined only on affine spaces that are included in $\mathbb{F}_2^s$ derived from the bent function g .

The Maiorana-McFarland construction [10], [6] generalises a family of bent functions introduced by Rothaus (referred to as Family R) [16]. In [4], this family is referred to as the Maiorana-McFarland primary construction. The author also discusses a secondary construction of the family, which is the focus of this work. The primary construction can be found in Dillon's thesis [6]. Conversely, the secondary construction was initially presented in [5], which pertains to a family of functions that exhibit correlation immunity. Subsequently, this secondary construction was adapted by the original concatenation principle to create a family of bent functions [3].

Additionally, the secondary construction discussed in [5] is also referenced in [4] as the class MM_r . It is important to note that within this class, the resulting functions do not need to be bent. The functions produced in this study belong to the class MM_r .

The first family of MM_r functions we obtain is derived directly from the family of bent functions outlined in the secondary construction presented in [3] and [4]. The second group of functions is obtained by iterating the first result for a specific case; these functions are also bent and belong to MM_r . However, they are not derived from the conditions that must be satisfied by the functions included in the secondary form of bent functions as stated in Proposition 79 of [4].

We utilised the Maiorana-McFarland secondary construction [3], [4] to construct bent functions. We start with an initial bent function $g : \mathbb{F}_2^{s-r} \rightarrow \mathbb{F}_2$. Next, we create 2^r affine spaces as subsets of $\mathbb{F}_2^s$, where we construct bent functions, which are extensions of the original function g . By considering a specific function $\phi : \mathbb{F}_2^s \rightarrow \mathbb{F}_2^r$ and using the previously defined concepts, we then derive a new bent function on $\mathbb{F}_2^{s+r}$. These bent functions extend the case where $r = 1$, as discussed in [9]. In that paper, a bent function defined on $\mathbb{F}_2^{s-1}$ is examined, resulting in the derivation of a bent function on $\mathbb{F}_2^{s+1}$. Two affine subspaces are considered: one consisting of vectors with even Hamming weight and the other with odd Hamming weight. In this work, we explore the general case where we start with a bent function on $\mathbb{F}_2^{s-r}$ and derive a bent function on $\mathbb{F}_2^{s+r}$. This introduces considerable complexity, as there are 2^r affine spaces involved in each of the demonstrations and additional conditions that must be considered. Furthermore, we present additional demonstrations comparing our general case to the specific case discussed in [9].

The first point addressed is the demonstration that we have created a partition of $\mathbb{F}_2^s$ consisting of the defined affine spaces (Proposition 1). This differs from the more straightforward scenario involving the two affine subspaces discussed in [9]. The balancedness of a bent function can be established by restricting it to elements with even Hamming weight or those with odd Hamming weight, as referenced in [9] and [2]. This approach allows us to generalize [2, Theorem V.2] to 2^r bent functions within their respective affine spaces (see Theorem 5). Thanks to this result, we can also extend the [9, Theorem 10]. Moreover, additional properties are necessary, along with a more thorough analysis, to establish the balancedness of the bent functions defined on the set of elements with even Hamming weight (see Theorem 7).

Additionally, we have derived a new formula to obtain bent functions, presented in Theorem 8, which arises from the repeated application of Theorem 6 in the case where $r = 1$. We discovered that this function maintains its balancedness when restricted to elements with even Hamming weight (see Corollary 1). Finally, to confirm that the functions described in Theorem 6 are distinct, we demonstrate this difference in Theorem 9. To support our findings, we present an algorithm that examines the case of $r = 2$ and an additional algorithm that applies the case of $r = 1$ twice.

2 Background

This chapter defines key concepts for this work, including Hamming distance, affine spaces, and non-linearity. The literature references these concepts, such as in [9], [11], [13], and [17].

A **Boolean function** is a function such that its images are 0 or 1. It is typically defined with domain $\mathbb{F}_2^n$. We denote the set of all Boolean functions $f : \mathbb{F}_2^n \rightarrow \mathbb{F}_2$ as $\mathcal{B}_n$.

Definition 1 Let $f, g \in \mathcal{B}_n$. The **Hamming distance** between f and g is denoted as $d_H(f, g)$, where $d_H(f, g) := \#\{\bar{x} \in \mathbb{F}_2^n \mid f(\bar{x}) \neq g(\bar{x})\}$.

Definition 2 The **algebraic normal form** (ANF) of a Boolean function $f : \mathbb{F}_2^n \rightarrow \mathbb{F}_2$ is expressed

$$f(\bar{x}) = \bigoplus_{\bar{u} \in \mathbb{F}_2^n} a_{\bar{u}} \bar{x}^{\bar{u}},$$

Definition 3 An **affine function** in $\mathcal{B}_n$ is defined as $l_{\bar{a}, a_0}(\bar{x}) := \bar{a} \cdot \bar{x} \oplus a_0$, where $\bar{a}, \bar{x} \in \mathbb{F}_2^n$ and $a_0 \in \mathbb{F}_2$. Here, $\bar{a} \cdot \bar{x} = a_1 x_1 \oplus a_2 x_2 \oplus \cdots \oplus a_n x_n$, with $\bar{a} = (a_1, a_2, \dots, a_n)$ and $\bar{x} = (x_1, x_2, \dots, x_n)$. The set of all affine functions is denoted by $\mathcal{A}_n$. If $a_0 = 0$, we refer to the function as $l_{\bar{a}}$ and call it a **linear function**.

The **non-linearity** of a Boolean function in $\mathcal{B}_n$, denoted as $Nl(f)$, measures its Hamming distance from the set of affine functions.

Theorem 1 *The non-linearity of a Boolean function $f \in \mathcal{B}_n$ is characterized by*

$$Nl(f) = 2^{n-1} - \frac{1}{2} \max_{\bar{a} \in \mathbb{F}_2^n} |\widehat{\mathcal{W}}_f(\bar{a})|,$$

where $\widehat{\mathcal{W}}_f(\bar{a}) := \sum_{\bar{x} \in \mathbb{F}_2^n} (-1)^{f(\bar{x}) \oplus \bar{a} \cdot \bar{x}}$, $\bar{a} \in \mathbb{F}_2^n$.

The function $\widehat{\mathcal{W}}_f$ is called the **Hadamard transform** of f .

Let us examine Boolean functions defined on an affine space $\mathcal{C} \subseteq \mathbb{F}_2^n$ as is mentioned in Proposition 1 of [3]. Let be a function $f : \mathcal{C} \subseteq \mathbb{F}_2^n \rightarrow \mathbb{F}_2$, then $Nl_{\mathcal{C}}(f) := \min_{l \in \mathcal{A}_n} d_H(f, l|_{\mathcal{C}})$ defines its non-linearity.

Similar to Theorem 1, but now focusing on affine spaces, non-linearity can be represented in the following manner:

Theorem 2 *Let $f : \mathcal{C} \subseteq \mathbb{F}_2^n \rightarrow \mathbb{F}_2$ be a function, where $\mathcal{C}$ has dimension m . Then, we have:*

$$Nl_{\mathcal{C}}(f) = 2^{m-1} - \frac{1}{2} \max_{\bar{a} \in \mathbb{F}_2^n} |\widehat{\mathcal{W}}_f(\bar{a})|,$$

where $\widehat{\mathcal{W}}_f(\bar{a}) = \sum_{\bar{x} \in \mathcal{C}} (-1)^{f(\bar{x}) \oplus \bar{a} \cdot \bar{x}}$.

The specific case where $\mathcal{C} = \mathbb{F}_2^n$ corresponds precisely to Theorem 1.

Boolean functions with maximum non-linearity are called **bent functions**.

Theorem 3 (Parseval's equation) *Let $f : \mathcal{C} \subseteq \mathbb{F}_2^n \rightarrow \mathbb{F}_2$ be a function, where $\mathcal{C}$ is of dimension m . Then,*

$$\sum_{\bar{a} \in \mathbb{F}_2^n} \widehat{\mathcal{W}}_f^2(\bar{a}) = 2^{m+n}.$$

Proof The expression on the left can be simplified as follows:

$$\sum_{\bar{a} \in \mathbb{F}_2^n} \widehat{\mathcal{W}}_f^2(\bar{a}) = \sum_{\bar{x}, \bar{y} \in \mathcal{C}} (-1)^{f(\bar{x}) \oplus f(\bar{y})} \sum_{\bar{a} \in \mathbb{F}_2^n} (-1)^{\bar{a} \cdot (\bar{x} \oplus \bar{y})}.$$

If $\bar{x} \neq \bar{y}$, the inner sum equals 0. However, if $\bar{x} = \bar{y}$, the inner sum equals 2^n .

We can obtain the desired result by counting the instances where $\bar{x} = \bar{y}$.

If $f : \mathcal{C} \subseteq \mathbb{F}_2^n \rightarrow \mathbb{F}_2$ is a bent function, then $\widehat{\mathcal{W}}_f(\bar{a}) = \pm 2^{m/2}$ for all $\bar{a} \in \mathbb{F}_2^n$.

The following theorem pertains to a specific class of bent functions known as the **Maiorana–McFarland secondary** construction.

Theorem 4 [3] [4] *Let the function $\phi(\bar{y}) : \mathbb{F}_2^s \rightarrow \mathbb{F}_2^r$ be defined such that for every $\bar{a} \in \mathbb{F}_2^r$, the preimage $\phi^{-1}(\bar{a})$ is an affine space of dimension $s - r$. Additionally, let there be a function $g_e(\bar{y}) : \mathbb{F}_2^s \rightarrow \mathbb{F}_2$, where $g_e|_{\phi^{-1}(\bar{a})}$ is a bent function. Then, the function*

$$f : \mathbb{F}_2^{r+s} \rightarrow \mathbb{F}_2, \quad (\bar{x}, \bar{y}) \mapsto \bar{x} \cdot \phi(\bar{y}) \oplus g_e(\bar{y}),$$

where $\bar{x} \in \mathbb{F}_2^r$, is a bent function.

It is important to note that if $n = r + s$, the existence of these functions is only guaranteed if $r \leq \frac{n}{2}$.

3 Constructing MM_r bent functions

In Theorem 4, if the functions ϕ and g_e are chosen arbitrarily within their specified domains and ranges, the resulting functions f will belong to the class MM_r [4]. As such, the class MM_r does not necessarily consist solely of bent functions. The objective of this section is to construct bent functions by employing the Maiorana-McFarland secondary construction, specifically focusing on bent MM_r functions derived from the conditions outlined in Theorem 4.

Reference [9] discusses the specific case where $r = 1$. This work will employ a similar approach to address the general case in which r can take any value. First, let $g : \mathbb{F}_2^{s-r} \rightarrow \mathbb{F}_2$ be a bent function. We will define a new function g_e .

Let $g_e|_{\mathcal{C}_i} := g_{e_i}$ for $i = 1, \dots, 2^r$, such that $g_{e_i} : \mathcal{C}_i \subset \mathbb{F}_2^s \rightarrow \mathbb{F}_2$, where $\mathcal{C}_i$ is an affine space. We define

$$g_{e_i}(\bar{x}_1 | \bar{x}_2) := g(\bar{x}_1),$$

with $\bar{x}_1 \in \mathbb{F}_2^{s-r}$ and $\bar{x}_2 \in \mathbb{F}_2^r$.

To ensure that g_e is well-defined, it is necessary to define the sets $\mathcal{C}_i$ for $i = 1, \dots, 2^r$ in such a way that they form a partition of $\mathbb{F}_2^s$.

In the scenario where $r = 1$ as mentioned in [9], the sets $\mathcal{C}_1$ and $\mathcal{C}_2$ represent the elements of even and odd Hamming weight in $\mathbb{F}_2^s$, respectively. We maintain this definition for $r = 1$. Now, we define the affine spaces for the general case. We arranged the elements of $\mathbb{F}_2^s$ in a specific way to understand the concepts better and demonstrate the obtained results.

To clarify the order in which elements from different sets, such as sets $\mathcal{A}_1$ and $\mathcal{B}_1$, are considered, these sets will be represented as matrices, where the rows correspond to the elements of each set.

Let $\mathcal{A}_1 \cup \mathcal{B}_1 = \mathbb{F}_2^s$, where $\mathcal{A}_1$ is the set of elements with even Hamming weight, and $\mathcal{B}_1$ is the set of elements with odd Hamming weight. For $0 < r < s$ and $i = 1, \dots, r$, we use the fundamental relations:

$$\begin{matrix} \mathcal{A}_{i+1} & \bar{0}_i \\ \mathcal{B}_{i+1} & \bar{1}_i \end{matrix} = \mathcal{A}_i \quad \text{and} \quad \begin{matrix} \mathcal{A}_{i+1} & \bar{1}_i \\ \mathcal{B}_{i+1} & \bar{0}_i \end{matrix} = \mathcal{B}_i, \quad i = 1, \dots, r.$$

Here, $\bar{0}_i = \begin{smallmatrix} 0 & 1 \\ \vdots & \vdots \\ 0 & 1 \end{smallmatrix}$ and $\bar{1}_i = \begin{smallmatrix} 1 & 0 \\ \vdots & \vdots \\ 1 & 0 \end{smallmatrix}$ are matrices with dimensions $2^{s-(i+1)} \times 1$.

Then, we can write a general case:

$$\mathbb{F}_2^s = \frac{\begin{smallmatrix} \mathcal{A}_1 & \mathcal{C}_{2^{r-1}} \\ \mathcal{B}_1 & \mathcal{C}_{2^{r-1}+1} \end{smallmatrix}}{\begin{smallmatrix} \mathcal{A}_{r+1} & \mathcal{C}_{2^r} \end{smallmatrix}} = \frac{\begin{smallmatrix} \bar{0}_r & \bar{X}_1 \\ \bar{1}_r & \bar{X}_1 \end{smallmatrix}}{\begin{smallmatrix} \bar{0}_r & \bar{X}_{2^{r-1}+1} \\ \bar{1}_r & \bar{X}_{2^{r-1}+1} \end{smallmatrix}} = \frac{\begin{smallmatrix} \bar{X}_{2^{r-1}} \\ \bar{X}_{2^r} \end{smallmatrix}}{\begin{smallmatrix} \bar{X}_{2^r} \end{smallmatrix}}.$$

The sets $\bar{X}_i$ for $i = 1, \dots, 2^r$ are considered as matrices with dimensions $2^{s-(r+1)} \times (r-1)$. All rows of $\bar{X}_i$ are equal, and for $i = 1, \dots, 2^{r-1}$, these rows represent all elements of $\mathbb{F}_2^{r-1}$ in the constructed order. The same applies for $i = 2^{r-1} + 1, \dots, 2^r$. For each i in the range $1, \dots, 2^r$, $\mathcal{C}_i$ is a matrix with dimensions $2^{s-r} \times s$, defined as follows:

$$\mathcal{C}_i = \begin{cases} \begin{smallmatrix} \mathcal{A}_{r+1} & \bar{0}_r & \bar{X}_i \\ \mathcal{B}_{r+1} & \bar{1}_r & \bar{X}_i \end{smallmatrix} & \text{if } i \text{ is odd} \\ \begin{smallmatrix} \mathcal{A}_{r+1} & \bar{1}_r & \bar{X}_i \\ \mathcal{B}_{r+1} & \bar{0}_r & \bar{X}_i \end{smallmatrix} & \text{if } i \text{ is even} \end{cases}$$

This formulation helps clarify how each $\mathcal{C}_i$ is constructed based on whether i is odd or even.

Remark 1 1. The set $\mathcal{A}_{r+1}$ is organized as a matrix with dimensions $2^{s-(r+1)} \times (s-r)$. It contains all the elements from $\mathbb{F}_2^{s-r}$ that have an even Hamming weight. In a similar way, the set $\mathcal{B}_{r+1}$ is also arranged in a matrix of the same dimensions, but it consists of all elements from $\mathbb{F}_2^{s-r}$ that have an odd Hamming weight.

2. We have the following representation:

$$\mathbb{F}_2^r = \frac{\begin{smallmatrix} \bar{0}_r & \bar{X}_1 \\ \bar{1}_r & \bar{X}_1 \end{smallmatrix}}{\begin{smallmatrix} \bar{1}_r & \bar{X}_{2^{r-1}} \\ \bar{0}_r & \bar{X}_{2^{r-1}} \end{smallmatrix}} = \frac{\begin{smallmatrix} \bar{0}_r & \bar{X}_{2^{r-1}+1} \\ \bar{1}_r & \bar{X}_{2^{r-1}+1} \end{smallmatrix}}{\begin{smallmatrix} \bar{1}_r & \bar{X}_{2^r} \\ \bar{0}_r & \bar{X}_{2^r} \end{smallmatrix}}.$$

3. For $i = 1, \dots, 2^{r-1}$, the elements of $\bar{X}_i$ have the following properties regarding their Hamming weights: if i is odd, the Hamming weight is even, and if i is even, the Hamming weight is odd.
 For $i = 2^{r-1} + 1, \dots, 2^r$, the elements of $\bar{X}_i$ exhibit the opposite behavior: if i is odd, the Hamming weight is odd, and if i is even, the Hamming weight is even.

Let's illustrate using the parameters $s = 5$ and $r = 3$ to clarify the general case and definitions. Consider $\mathbb{F}_2^5 = \mathcal{A}_1 \cup \mathcal{B}_1$.

			$\mathcal{A}_4 \bar{0}_3 \bar{0}_2$		
			$\mathcal{B}_4 \bar{1}_3 \bar{0}_2$		
			— — — — $\bar{0}_1$	00000	00001
	$\mathcal{C}_1$		$\mathcal{A}_4 \bar{1}_3 \bar{1}_2$	11000	11001
			$\mathcal{B}_4 \bar{0}_3 \bar{1}_2$	10100	10101
	— —		— — — —	01100	01101
	$\mathcal{C}_2$		$\mathcal{A}_4 \bar{0}_3 \bar{1}_2$	00110	00111
	— —		$\mathcal{B}_4 \bar{1}_3 \bar{1}_2$	11110	11111
	$\mathcal{C}_3$		— — — — $\bar{1}_1$	10010	10011
	— —		$\mathcal{A}_4 \bar{1}_3 \bar{0}_2$	01010	01011
$\mathbb{F}_2^5 =$	$\mathcal{A}_1$	$\mathcal{C}_4$	$\mathcal{B}_4 \bar{0}_3 \bar{0}_2$		
	— —		— — — —	$\mathcal{A}_1 =$ — — —	$\mathcal{B}_1 =$ — — —
	$\mathcal{B}_1$	$\mathcal{C}_5$	$\mathcal{A}_4 \bar{0}_3 \bar{0}_2$	00011	00010
		— —	$\mathcal{B}_4 \bar{1}_3 \bar{0}_2$	11011	11010
		$\mathcal{C}_6$	— — — — $\bar{1}_1$	10111	10110
	— —		$\mathcal{A}_4 \bar{1}_3 \bar{1}_2$	01111	01110
	$\mathcal{C}_7$		$\mathcal{B}_4 \bar{0}_3 \bar{1}_2$	00101	00100
	— —		— — — —	11101	11100
	$\mathcal{C}_8$		$\mathcal{A}_4 \bar{0}_3 \bar{1}_2$	10001	10000
			$\mathcal{B}_4 \bar{1}_3 \bar{1}_2$	01001	01000
			— — — — $\bar{0}_1$		
			$\mathcal{A}_4 \bar{1}_3 \bar{0}_2$		
			$\mathcal{B}_4 \bar{0}_3 \bar{0}_2$		

$$\mathcal{C}_1 = \begin{matrix} 00000 \\ 11000 \\ 10100 \\ 01100 \end{matrix} \quad \mathcal{C}_2 = \begin{matrix} 00110 \\ 11110 \\ 10010 \\ 01010 \end{matrix} \quad \mathcal{C}_3 = \begin{matrix} 00011 \\ 11011 \\ 10111 \\ 01111 \end{matrix} \quad \mathcal{C}_4 = \begin{matrix} 00101 \\ 11101 \\ 10001 \\ 01001 \end{matrix}$$

$$\mathcal{C}_5 = \begin{matrix} 00001 \\ 11001 \\ 10101 \\ 01101 \end{matrix} \quad \mathcal{C}_6 = \begin{matrix} 00111 \\ 11111 \\ 10011 \\ 01011 \end{matrix} \quad \mathcal{C}_7 = \begin{matrix} 00010 \\ 11010 \\ 10110 \\ 01110 \end{matrix} \quad \mathcal{C}_8 = \begin{matrix} 00100 \\ 11100 \\ 10000 \\ 01000 \end{matrix}$$

$$\mathcal{A}_4 = \begin{matrix} 00 \\ 11 \end{matrix} \quad \mathcal{B}_4 = \begin{matrix} 10 \\ 01 \end{matrix} \quad \bar{0}_3 = \begin{matrix} 0 \\ 0 \end{matrix} \quad \bar{1}_3 = \begin{matrix} 1 \\ 1 \end{matrix}$$

$\mathcal{C}_i \subset \mathbb{F}_2^5$, $i = 1, \dots, 8$, $\mathcal{A}_4 \cup \mathcal{B}_4 = \mathbb{F}_2^2$.

$\mathcal{A}_4$ is the set of elements in $\mathbb{F}_2^2$ with even Hamming weight. $\mathcal{B}_4$ is the set of elements in $\mathbb{F}_2^2$ with odd Hamming weight.

To establish a clear definition for g_e and g_{e_i} for $i = 1, \dots, 2^r$:

Proposition 1 *The set $\{\mathcal{C}_1, \dots, \mathcal{C}_{2^r}\}$ is a partition of $\mathbb{F}_2^s$ and each element is an affine space.*

Proof We know that

1. For $i = 1, \dots, 2^r$:
If i is odd,

$$a) \quad \mathcal{C}_i = \begin{array}{c} \mathcal{A}_{r+1} \quad \begin{array}{c} 0, x_{i(s-r+2)}, \dots, x_{is} \\ \vdots \\ 0, x_{i(s-r+2)}, \dots, x_{is} \end{array} \\ \hline \mathcal{B}_{r+1} \quad \begin{array}{c} 1, x_{i(s-r+2)}, \dots, x_{is} \\ \vdots \\ 1, x_{i(s-r+2)}, \dots, x_{is} \end{array} \end{array}.$$

In particular, $\mathcal{C}_1$ is a vector space of dimension $s-r$, where $(x_{1(s-r+2)}, \dots, x_{1s}) = (0, \dots, 0)$.

If i is even,

$$b) \quad \mathcal{C}_i = \begin{array}{c} \mathcal{A}_{r+1} \quad \begin{array}{c} 1, x_{i(s-r+2)}, \dots, x_{is} \\ \vdots \\ 1, x_{i(s-r+2)}, \dots, x_{is} \end{array} \\ \hline \mathcal{B}_{r+1} \quad \begin{array}{c} 0, x_{i(s-r+2)}, \dots, x_{is} \\ \vdots \\ 0, x_{i(s-r+2)}, \dots, x_{is} \end{array} \end{array}.$$

Now, we can express $\mathcal{C}_i$ for $i = 1, \dots, 2^r$ as affine spaces:

Let $\mathcal{C}_i$ be of the form $a)$ and let $\bar{x}_i = (\bar{x}'_i, 0, x_{i(s-r+2)}, \dots, x_{is}) \in \mathcal{C}_i$, where $\bar{x}'_i \in \mathcal{A}_{r+1}$ is a fixed element. Then we have:

$$\bar{x}_i + \mathcal{C}_1 = \mathcal{C}_i.$$

Let $\mathcal{C}_i$ be of the form $b)$ and let $\bar{x}_i = (\bar{x}'_i, 1, x_{i(s-r+2)}, \dots, x_{is})$, with $\bar{x}'_i \in \mathcal{A}_{r+1}$ as a fixed element. Then we also have:

$$\bar{x}_i + \mathcal{C}_1 = \mathcal{C}_i.$$

Additionally, the sets $\mathcal{C}_i$ for $i = 1, \dots, 2^r$ are distinct and their intersections are empty. We will divide the demonstration into two cases: when $\mathcal{C}_i$ for $i \in$

$\{1, \dots, 2^{r-1}\}$ has an empty intersection with the other affine spaces, and when $\mathcal{C}_i$ for $i \in \{2^{r-1} + 1, \dots, 2^r\}$ has an empty intersection.

In the first case, let $\bar{x}_i \in \mathcal{C}_i$ for $i \in \{1, \dots, 2^{r-1}\}$, with $\bar{x}_i = (\bar{x}'_i, k, \bar{x}''_i)$, where $k = 0$ or $k = 1$, $\bar{x}'_i \in \mathcal{A}_{r+1}$ or $\bar{x}'_i \in \mathcal{B}_{r+1}$, and $\bar{x}''_i$ is a row of $\bar{X}_i$.

We claim that $\bar{x}_i \notin \mathcal{C}_j$ for $j \neq i$ with $j = 1, \dots, 2^{r-1}$, because $\bar{x}''_i$ is not a row of $\bar{X}_j$. Additionally, $\bar{x}_i \notin \mathcal{C}_j$ for $j \neq i$ with $j = 2^{r-1} + 1, \dots, 2^r$ since, in the unique case where $\bar{x}''_i = \bar{x}''_j$, we have $j = 2^r - i + 1$. In this scenario, if $\bar{x}'_i \in \mathcal{A}_{r+1}$, then $\bar{x}'_j \in \mathcal{B}_{r+1}$, and if $\bar{x}'_i \in \mathcal{B}_{r+1}$, then $\bar{x}'_j \in \mathcal{A}_{r+1}$. Thus, $\bar{x}_i \notin \mathcal{C}_j$.

In the other case, when $\bar{x}_i \in \mathcal{C}_i$ for $i \in \{2^{r-1} + 1, \dots, 2^r\}$, the proof follows a similar argument.

The union of the affine spaces is the set $\mathbb{F}_2^s$. Therefore, we have established the desired result.

Remark 2 [9]

1. Let $g : \mathbb{F}_2^{s-r} \rightarrow \mathbb{F}_2$ be a bent function such that $g|_{\mathcal{A}_{r+1}}$ is balanced.

- (a) If $\widehat{\mathcal{W}}_g(\bar{0}) = 2^{\frac{s-r}{2}}$, then
- $$|g|_{\mathcal{A}_{r+1}}^{-1}(0)| = 2^{s-r-2}$$
- $$|g|_{\mathcal{A}_{r+1}}^{-1}(1)| = 2^{s-r-2}$$
- $$|g|_{\mathcal{B}_{r+1}}^{-1}(0)| = 2^{s-r-2} + 2^{\frac{s-r-2}{2}}$$
- $$|g|_{\mathcal{B}_{r+1}}^{-1}(1)| = 2^{s-r-2} - 2^{\frac{s-r-2}{2}}$$

- (b) If $\widehat{\mathcal{W}}_g(\bar{0}) = -2^{\frac{s-r}{2}}$, then
- $$|g|_{\mathcal{A}_{r+1}}^{-1}(0)| = 2^{s-r-2}$$
- $$|g|_{\mathcal{A}_{r+1}}^{-1}(1)| = 2^{s-r-2}$$
- $$|g|_{\mathcal{B}_{r+1}}^{-1}(0)| = 2^{s-r-2} - 2^{\frac{s-r-2}{2}}$$
- $$|g|_{\mathcal{B}_{r+1}}^{-1}(1)| = 2^{s-r-2} + 2^{\frac{s-r-2}{2}}$$

2. Similar observations if $g|_{\mathcal{B}_{r+1}}$ is balanced.

The claim that if g is a bent function, then $g|_{\mathcal{A}_{r+1}}$ is balanced or $g|_{\mathcal{B}_{r+1}}$ is balanced [9] is essential to proving the following theorem.

Using the notation accorded above.

Theorem 5 Let $g : \mathbb{F}_2^{s-r} \rightarrow \mathbb{F}_2$ be a bent function. Then, for $i = 1, \dots, 2^r$, the functions $g_{e_i} : \mathcal{C}_i \rightarrow \mathbb{F}_2$ are also bent functions, defined as follows:

$$g_{e_i}(\bar{x}_1 | \bar{x}_2) := g(\bar{x}_1),$$

where $g_{e_i} = g_{e_i}$, $\bar{x}_1 \in \mathbb{F}_2^{s-r}$, and $\bar{x}_2 \in \mathbb{F}_2^r$.

Proof Let $\bar{b} = (\bar{b}_1, b_{s-r+1}, \bar{b}_2) \in \mathbb{F}_2^s$, where $\bar{b}_1 \in \mathbb{F}_2^{s-r}$, $\bar{b}_2 \in \mathbb{F}_2^{r-1}$, and $b_{s-r+1} \in \mathbb{F}_2$. Let $\bar{x} = (\bar{x}_1, x_{s-r+1}, \bar{x}_2) \in \mathcal{C}_i$, where $\bar{x}_1 \in \mathbb{F}_2^{s-r}$, $\bar{x}_2 \in \mathbb{F}_2^{r-1}$, $x_{s-r+1} \in \mathbb{F}_2$, and $i \in \{1, \dots, 2^r\}$.

We aim to prove that $g_{e_i} : \mathcal{C}_i \rightarrow \mathbb{F}_2$ is a bent function.

If $\bar{x} = (\bar{x}_1, x_{s-r+1}, \bar{x}_2) \in \mathcal{C}_i$, then $\bar{x}_2 = \bar{c}_2$, where $\bar{c}_2$ is a constant vector.

$$\widehat{\mathcal{W}}_{g_{e_i}}(\bar{b}) = \sum_{\bar{x} \in \mathcal{C}_i} (-1)^{g_{e_i}(\bar{x}) + \bar{x} \cdot \bar{b}} = \sum_{\bar{x} \in \mathcal{C}_i} (-1)^{g_{e_i}(\bar{x}) + \bar{x}_1 \cdot \bar{b}_1 + x_{s-r+1} b_{s-r+1} + \bar{c}_2 \cdot \bar{b}_2}.$$

Case 1: If $b_{s-r+1} = 0$:

$$\widehat{\mathcal{W}}_{g_{e_i}}(\bar{b}) = (-1)^{\bar{c}_2 \cdot \bar{b}_2} \sum_{\bar{x}_1 \in \mathbb{F}_2^{s-r}} (-1)^{g(\bar{x}_1) + \bar{x}_1 \cdot \bar{b}_1} = (-1)^{\bar{c}_2 \cdot \bar{b}_2} \widehat{\mathcal{W}}_g(\bar{b}_1).$$

Case 2: If $b_{s-r+1} = 1$ and i is odd:

$$\widehat{\mathcal{W}}_{g_{e_i}}(\bar{b}) = (-1)^{\bar{c}_2 \cdot \bar{b}_2} \sum_{\bar{x} \in \mathcal{C}_i} \left((-1)^{g_{e_i}(\bar{x}_1, 0, \bar{x}_2) + \bar{x}_1 \cdot \bar{b}_1} + (-1)^{g_{e_i}(\bar{x}_1, 1, \bar{x}_2) + \bar{x}_1 \cdot \bar{b}_1 + 1} \right).$$

This simplifies to:

$$(-1)^{\bar{c}_2 \cdot \bar{b}_2} \sum_{\bar{x}_1 \in \mathcal{A}_{r+1}} (-1)^{g_{e_i}(\bar{x}_1, 0, \bar{x}_2) + \bar{x}_1 \cdot \bar{b}_1} + (-1)^{\bar{c}_2 \cdot \bar{b}_2 + 1} \sum_{\bar{x}_1 \in \mathcal{B}_{r+1}} (-1)^{g_{e_i}(\bar{x}_1, 1, \bar{x}_2) + \bar{x}_1 \cdot \bar{b}_1}.$$

Case 3: If $b_{s-r+1} = 1$ and i is even:

$$\widehat{\mathcal{W}}_{g_{e_i}}(\bar{b}) = (-1)^{\bar{c}_2 \cdot \bar{b}_2} \sum_{\bar{x}_1 \in \mathcal{B}_{r+1}} (-1)^{g_{e_i}(\bar{x}_1, 0, \bar{x}_2) + \bar{x}_1 \cdot \bar{b}_1} + (-1)^{\bar{c}_2 \cdot \bar{b}_2 + 1} \sum_{\bar{x}_1 \in \mathcal{A}_{r+1}} (-1)^{g_{e_i}(\bar{x}_1, 1, \bar{x}_2) + \bar{x}_1 \cdot \bar{b}_1}.$$

In both cases where $b_{s-r+1} = 1$, as $(g + l_{\bar{b}_1})|_{\mathcal{A}_{r+1}}$ is balanced or $(g + l_{\bar{b}_1})|_{\mathcal{B}_{r+1}}$ is balanced, then we have:

$$\widehat{\mathcal{W}}_{g_{e_i}}(\bar{b}) = \pm \widehat{\mathcal{W}}_g(\bar{b}_1).$$

Since g is a bent function, it follows that g_{e_i} is also a bent function.

We use the addition symbol $+$ for exponent operations instead of the symbol $\oplus$, as the results are identical.

To define all elements of Theorem 4, we express the function $\phi : \mathbb{F}_2^s \rightarrow \mathbb{F}_2^r$ in terms of its inverse images:

$$\mathcal{C}_1 := \phi^{-1}(0, 0, \dots, 0), \quad \mathcal{C}_2 := \phi^{-1}(0, 0, \dots, 1), \quad \dots, \quad \mathcal{C}_{2^{r-1}} := \phi^{-1}(0, 1, \dots, 1),$$

$$\mathcal{C}_{2^{r-1}+1} := \phi^{-1}(1, 0, \dots, 0), \quad \dots, \quad \mathcal{C}_{2^r} := \phi^{-1}(1, 1, \dots, 1).$$

The images can be considered in any order regarding the affine spaces. For clarity, we illustrate the definition using lexicographic order. In the specific case where $r = 1$, as discussed in [9], we have: $\mathcal{C}_1 = \phi^{-1}(0)$ and $\mathcal{C}_2 = \phi^{-1}(1)$. We maintain this relationship for $r = 1$.

We can apply Theorem 4 to obtain bent functions using the previous definitions.

Theorem 6 Let $g : \mathbb{F}_2^{s-r} \rightarrow \mathbb{F}_2$ be a bent function, and let g_e and ϕ be defined as mentioned above. Then, the function

$$f : \mathbb{F}_2^{r+s} \rightarrow \mathbb{F}_2, (\bar{x}, \bar{y}) \mapsto \bar{x} \cdot \phi(\bar{y}) \oplus g_e(\bar{y}),$$

where $\bar{x} \in \mathbb{F}_2^r$ and $\bar{y} \in \mathbb{F}_2^s$, is a bent function.

Proof By the definitions of ϕ , g_e , and Proposition 1, along with Theorem 5, we gather all elements that satisfy the conditions of Theorem 4. Thus, f is a bent function.

We aim to achieve a specific level of balancedness in the domain of the bent functions we obtained. Therefore, we will make observations and provide additional conditions in the following theorem.

Remark 3 Let $\mathcal{A} := \{\bar{x} \in \mathbb{F}_2^n \mid w_H(\bar{x}) \text{ is even}\}$ and $\mathcal{B} := \{\bar{x} \in \mathbb{F}_2^n \mid w_H(\bar{x}) \text{ is odd}\}$. Consider $\bar{a}, \bar{1} \in \mathbb{F}_2^n$, where $\bar{1} = (1, \dots, 1)$.

1. If $w_H(\bar{a}) \neq n$ and $w_H(\bar{a}) \neq 0$, then $l_{\bar{a}|\mathcal{A}}$ is balanced, and $l_{\bar{a}|\mathcal{B}}$ is balanced.
2. If n is even and $\bar{a} = \bar{1}$, then $l_{\bar{a}}(\mathcal{A}) = \{0\}$. If n is odd and $\bar{a} = \bar{1}$, then $l_{\bar{a}}(\mathcal{B}) = \{1\}$.

In the following theorem, we will consider the sets $\mathcal{C}_i$ for $i = 1, \dots, 2^r$, as well as the sets $\mathcal{A}_1$, $\mathcal{B}_1$, $\mathcal{A}_r$, and $\mathcal{B}_r$ that are defined at the beginning of this section.

Theorem 7 Let r be an odd integer, and let $g : \mathbb{F}_2^{s-r} \rightarrow \mathbb{F}_2$ be a bent function. Define g_e as specified above. Additionally, let $\phi(\mathcal{C}_i)$ be an even image for $i = 1, \dots, 2^{r-1}$ and an odd image for $i = 2^{r-1} + 1, \dots, 2^r$. Then, the bent function

$$f : \mathbb{F}_2^{r+s} \rightarrow \mathbb{F}_2, (\bar{x}, \bar{y}) \mapsto \bar{x} \cdot \phi(\bar{y}) \oplus g_e(\bar{y}), \quad \bar{x} \in \mathbb{F}_2^r, \quad \bar{y} \in \mathbb{F}_2^s,$$

satisfies that the restriction of f to the set $\{(\bar{x}, \bar{y}) \mid w_H((\bar{x}, \bar{y})) \text{ is even}\}$ is balanced.

Proof Consider, without loss of generality, the case where $g|_{\mathcal{A}_{r+1}}$ is balanced and $\widehat{\mathcal{W}}_g(\bar{0}) = 2^{\frac{s-r}{2}}$.

Since we aim to determine the balancedness in $\mathbb{F}_2^{r+s}$, restricted to the set of vectors of even Hamming weight, we will examine the following necessary subcases:

1. $\bar{x}$ is even, and $\bar{y} \in \mathcal{C}_i$, where $i = 1, \dots, 2^{r-1}$.
2. $\bar{x}$ is odd, and $\bar{y} \in \mathcal{C}_i$, where $i = 2^{r-1} + 1, \dots, 2^r$.

In the first subcase, we know that $\phi(\mathcal{C}_i)$ has a constant even image, and $g_e|_{\mathcal{C}_i} = g_e$ has $2^{s-r-2} + 2^{s-r-2} + 2^{\frac{s-r-2}{2}}$ images equal to zero (see Remark 2). For each $\bar{x} \in \mathbb{F}_2^r$, depending on whether $\bar{x} \cdot \phi(\mathcal{C}_i)$ is zero or one, we have:

- If $\bar{x} \cdot \phi(\mathcal{C}_i) = 0$, then $f(\bar{x}, \mathcal{C}_i) = \bar{x} \cdot \phi(\mathcal{C}_i) \oplus g_e(\mathcal{C}_i)$ has $2^{s-r-2} + 2^{s-r-2} + 2^{\frac{s-r-2}{2}}$ images equal to zero.

- If $\bar{x} \cdot \phi(C_i) = 1$, then $f(\bar{x}, C_i)$ has $2^{s-r-2} + 2^{s-r-2} - 2^{\frac{s-r-2}{2}}$ images equal to zero.

On the other hand, $\bar{y} \mapsto \bar{x} \cdot \phi(\bar{y})$ is balanced in $\mathcal{A}_1$ if $\bar{x} \neq \bar{0}$, given that $\bar{z} \mapsto \bar{x} \cdot \bar{z}$ is balanced when $\bar{z}$ is restricted to the set of vectors with even Hamming weight in $\mathbb{F}_2^r$ (see Remark 3, item 1). Since there are $2^{r-1}-1$ elements $\bar{x} \in \mathbb{F}_2^r$ with even Hamming weight, distinct from the zero vector, the total number of images equal to zero is

$$(2^{r-1}-1) \left[2^{r-2} \left((2^{s-r-2} + 2^{s-r-2} + 2^{\frac{s-r-2}{2}}) + (2^{s-r-2} + 2^{s-r-2} - 2^{\frac{s-r-2}{2}}) \right) \right].$$

Moreover, $\bar{0} \cdot \phi(\bar{y}) = 0$ for all $\bar{y} \in \mathcal{A}_1$. Thus, we have an additional

$$2^{r-1}(2^{s-r-2} + 2^{s-r-2} + 2^{\frac{s-r-2}{2}})$$

images equal to zero.

The second subcase is similar, but $\bar{y} \mapsto \bar{x} \cdot \phi(\bar{y})$ is balanced in $\mathcal{B}_1$ if $w_H(\bar{x}) \neq n$. Since there are $2^{r-1}-1$ elements $\bar{x} \in \mathbb{F}_2^r$ with odd Hamming weight, the total number of images equal to zero is

$$(2^{r-1}-1) \left[2^{r-2} \left((2^{s-r-2} + 2^{s-r-2} + 2^{\frac{s-r-2}{2}}) + (2^{s-r-2} + 2^{s-r-2} - 2^{\frac{s-r-2}{2}}) \right) \right].$$

Additionally, $\bar{1} \cdot \phi(\bar{y}) = 1$ for all $\bar{y} \in \mathcal{B}_1$, as noted in Remark 3, item 2. Therefore, we also have

$$2^{r-1}(2^{s-r-2} + 2^{s-r-2} - 2^{\frac{s-r-2}{2}})$$

images equal to zero.

Finally, by summing all the images equal to zero, we find that the total is 2^{s+r-2} . Thus, f is balanced when restricted to the set of vectors with even Hamming weight.

By considering other cases and following a similar analysis, we can conclude that we obtain the same result.

The previous results generalize the case $r = 1$ discussed in [9]. Now, we aim to establish a relationship between the specific case $r = 1$ and the general case r .

4 Applying MM_1 many times

Suppose we have a bent function $g : \mathbb{F}_2^{s-l} \rightarrow \mathbb{F}_2$, where $\bar{y} \mapsto g(\bar{y})$, and we want to construct a bent function over $\mathbb{F}_2^{l+s}$ by applying Theorem 6 l times, using the specific case where $r = 1$. This method is outlined in Algorithm 1 of [9]; however, it does not provide a specific expression for the new function.

First, we define the function ϕ'_1 based on the case $r = 1$ as described in Theorem 6 over $\mathbb{F}_2^{s-l}$. We then repeat this process l times, resulting in l functions:

$$\phi'_1 : \mathbb{F}_2^{s-l+2(1)-1} \rightarrow \mathbb{F}_2, \quad \phi'_2 : \mathbb{F}_2^{s-l+2(2)-1} \rightarrow \mathbb{F}_2, \quad \dots, \quad \phi'_l : \mathbb{F}_2^{s-l+2(l)-1} \rightarrow \mathbb{F}_2.$$

Next, we can define a function $\phi' : \mathbb{F}_2^{s+l-1} \rightarrow \mathbb{F}_2^l$ as follows:

$$\begin{aligned} & \phi'(x_2, \dots, x_l, \bar{y}_1, \bar{y}_2) \\ &:= [\phi'_1(x_2, \dots, x_l, \bar{y}_1, \bar{y}_2), \dots, \phi'_l(x_l, \bar{y}_1, y_{s-l+1}, y_{s-l+2}), \phi'_1(\bar{y}_1, y_{s-l+1})], \end{aligned}$$

where $\bar{y}_1 \in \mathbb{F}_2^{s-l}$, $\bar{y}_2 = (y_{s-l+1}, \dots, y_s) \in \mathbb{F}_2^l$, and $x_2, \dots, x_l \in \mathbb{F}_2$.

Notice that if we consider a constant $\bar{c} = (c_2, \dots, c_l)$, then for all $(\bar{y}_1, \bar{y}_2) \in \mathcal{C}_i$ for each $i \in \{1, \dots, 2^l\}$, the outputs of $\phi'(\bar{c}, \bar{y}_1, \bar{y}_2)$ are equal. Furthermore, the set of outputs from $\phi'(\bar{c}, \bar{y}_1, \bar{y}_2)$ will include all the elements of $\mathbb{F}_2^l$, each repeated 2^{s-l} times, preserving the initial order accorded to $\mathbb{F}_2^s$.

More details in the next result.

Theorem 8 *Let $g : \mathbb{F}_2^{s-l} \rightarrow \mathbb{F}_2$, defined by $\bar{y}_1 \mapsto g(\bar{y}_1)$, be a bent function. Let ϕ' be defined as mentioned earlier. Then, there exists a bent function*

$$g_l : \mathbb{F}_2^{l+s} \rightarrow \mathbb{F}_2, \quad (\bar{x}, \bar{y}_1, \bar{y}_2) \mapsto \bar{x} \cdot \phi'(x_2, \dots, x_l, \bar{y}_1, \bar{y}_2) \oplus g_e(\bar{y}_1, \bar{y}_2),$$

where $g_e(\bar{y}_1, \bar{y}_2) = g(\bar{y}_1)$ for all $\bar{y}_1 \in \mathbb{F}_2^{s-l}$ and $\bar{y}_2 \in \mathbb{F}_2^l$. Here, $\bar{x} = (x_1, x_2, \dots, x_l) \in \mathbb{F}_2^l$.

Proof We will consider the case where $r = 1$ for l times.

Step 1. Let $g : \mathbb{F}_2^{s-l} \rightarrow \mathbb{F}_2$ be a bent function defined by $\bar{y}_1 \mapsto g(\bar{y}_1)$. We then define $g_e : \mathbb{F}_2^{s-l+1} \rightarrow \mathbb{F}_2$ and $\phi'_1 : \mathbb{F}_2^{s-l+1} \rightarrow \mathbb{F}_2$ as specified in Theorem 6 (in the case where $r = 1$) to construct a bent function $g_1 : \mathbb{F}_2^{s-l+2} \rightarrow \mathbb{F}_2$. We have:

$$g_1(x_l, \bar{\mathbf{y}}_1, y_{s-l+1}) = x_l \phi'_1(\bar{\mathbf{y}}_1, y_{s-l+1}) + g_e(\bar{\mathbf{y}}_1, y_{s-l+1}).$$

Step 2. Following a similar approach to Step 1, we now take the bent function $g_1 : \mathbb{F}_2^{s-l+2} \rightarrow \mathbb{F}_2$ and define $g_{1e} : \mathbb{F}_2^{s-l+3} \rightarrow \mathbb{F}_2$ and $\phi'_2 : \mathbb{F}_2^{s-l+3} \rightarrow \mathbb{F}_2$ as in Theorem 6. This leads us to a new bent function $g_2 : \mathbb{F}_2^{s-l+4} \rightarrow \mathbb{F}_2$. We can express this function as follows:

$$\begin{aligned} & g_2(x_{l-1}, \mathbf{x}_l, \bar{\mathbf{y}}_1, \mathbf{y}_{s-l+1}, y_{s-l+2}) \\ &= x_{l-1} \phi'_2(\mathbf{x}_l, \bar{\mathbf{y}}_1, \mathbf{y}_{s-l+1}, y_{s-l+2}) + g_{1e}(\mathbf{x}_l, \bar{\mathbf{y}}_1, \mathbf{y}_{s-l+1}, y_{s-l+2}) \\ &= x_{l-1} \phi'_2(x_l, \bar{y}_1, y_{s-l+1}, y_{s-l+2}) + x_l \phi'_1(\bar{y}_1, y_{s-l+1}) + g_e(\bar{y}_1, y_{s-l+1}) \\ &= (x_{l-1}, x_l) \cdot [\phi'_2(x_l, \bar{y}_1, y_{s-l+1}, y_{s-l+2}), \phi'_1(\bar{y}_1, y_{s-l+1})] + g_e(\bar{y}_1, y_{s-l+1}). \end{aligned}$$

Note that the relationship from Step 1 gives us the second equality here.

Step 3. Continuing in this manner for l times, we can construct a bent function $g_l : \mathbb{F}_2^{s-l+2l} \rightarrow \mathbb{F}_2$:

$$\begin{aligned} & g_l(x_1, \mathbf{x}_2, \dots, \mathbf{x}_{l-1}, \mathbf{x}_l, \bar{\mathbf{y}}_1, \mathbf{y}_{s-l+1}, \mathbf{y}_{s-l+2}, \dots, \mathbf{y}_{s-l+l-1}, y_{s-l+l}) \\ &= \bar{x} \cdot [\phi'_l(x_2, \dots, x_l, \bar{y}_1, \bar{y}_2), \dots, \phi'_2(x_l, \bar{y}_1, y_{s-l+1}, y_{s-l+2}), \phi'_1(\bar{y}_1, y_{s-l+1})] \\ &\quad + g_e(\bar{y}_1, y_{s-l+1}) \\ &= \bar{x} \cdot \phi'(x_2, \dots, x_l, \bar{y}_1, \bar{y}_2) + g_e(\bar{y}_1, \bar{y}_2). \end{aligned}$$

It is important to note that we have used the extension of g , denoted g_e , interchangeably with $r = 1$ and $r = l$.

Therefore, we conclude that:

$$g_l(\bar{x}, \bar{y}_1, \bar{y}_2) = \bar{x} \cdot \phi'(x_2, \dots, x_l, \bar{y}_1, \bar{y}_2) + g_e(\bar{y}_1, \bar{y}_2).$$

It is important to note that the bent function g_l derived from Theorem 8 can be expressed in the form MM_r , where $1 \leq r \leq l$. However, we cannot assert that the functions ϕ and g_e , which correspond to the MM_r form, satisfy the conditions outlined in Theorem 4.

In the following result, we do not require r to be an odd integer in order to achieve balancedness within the domain, specifically when restricted to vectors of even Hamming weight. Additionally, we consider ϕ' as stated in the Theorem 8.

Corollary 1 *Let $g : \mathbb{F}_2^{s-r} \rightarrow \mathbb{F}_2$ be a bent function, with g_e and ϕ' defined as previously described. Then, the bent function*

$$g_r : \mathbb{F}_2^{r+s} \rightarrow \mathbb{F}_2, (\bar{x}, \bar{y}_1, \bar{y}_2) \mapsto \bar{x} \cdot \phi'(x_2, \dots, x_r, \bar{y}_1, \bar{y}_2) \oplus g_e(\bar{y}_1, \bar{y}_2)$$

will satisfy that $f_{\{(\bar{x}, \bar{y}_1, \bar{y}_2) | w_H((\bar{x}, \bar{y}_1, \bar{y}_2)) \text{ is even}\}}$ is balanced. Here, $g_e(\bar{y}_1, \bar{y}_2) = g(\bar{y}_1)$ for all $\bar{y}_1 \in \mathbb{F}_2^{s-r}$, $\bar{x} \in \mathbb{F}_2^r$, $\bar{y}_2 \in \mathbb{F}_2^r$, with $\bar{x} = (x_1, x_2, \dots, x_r)$ and $\bar{y}_2 = (y_{s-r+1}, \dots, y_s)$.

Proof The proof is straightforward. We utilise Theorem 8 multiple times (particularly for the case when $l = 1$). Each instance involves constructing a balanced bent function restricted to the set of vectors with even Hamming weight in its domain [9].

5 Comparing constructions

We have obtained a similar expression considering the case $r = l$ and the case $r = 1$ l times. However, the obtained functions are different. In the following result, we demonstrate this claim. In this way, it is possible to construct more instances of bent functions.

Theorem 9 *Let f be the bent function obtained using Theorem 6 when $r = l$, and let g_l be the bent function obtained using Theorem 8, $r = 1$ l times. Then, $f \neq g_l$.*

Proof Let ϕ be defined as Theorem 6, and let $\phi' = (\phi'_1, \dots, \phi'_l)$ be as stated in Theorem 8. Note that the function ϕ' includes the variables $x_2, \dots, x_l$ in its domain, and we can express $\phi = (\phi_1, \dots, \phi_l)$, where $\phi_1, \dots, \phi_l$ represent the component functions.

Suppose $\phi_l(C_i) = \{0\}$ for some $i \in \{1, \dots, 2^l \mid i \text{ even}\}$. Then for all pairs $(\bar{y}_1, \bar{y}_2) \in C_i$ with $\bar{y}_1 \in \mathbb{F}_2^{s-l}$ and $\bar{y}_2 \in \mathbb{F}_2^l$, we have:

$$\begin{aligned} f(0, \dots, 0, 1, \bar{y}_1, \bar{y}_2) &= (0, \dots, 0, 1) \cdot \phi(\bar{y}_1, \bar{y}_2) \oplus g_e(\bar{y}_1, \bar{y}_2) \\ &= \phi_l(\bar{y}_1, \bar{y}_2) \oplus g_e(\bar{y}_1, \bar{y}_2) = g_e(\bar{y}_1, \bar{y}_2). \end{aligned}$$

On the other hand,

$$\begin{aligned} g_l(0, \dots, 0, 1, \bar{y}_1, \bar{y}_2) &= (0, \dots, 0, 1) \cdot \phi'(0, \dots, 0, 1, \bar{y}_1, \bar{y}_2) \oplus g_e(\bar{y}_1, \bar{y}_2) \\ &= \phi'_1(\bar{y}_1, y_{s-l+1}) + g_e(\bar{y}_1, \bar{y}_2) = 1 \oplus g_e(\bar{y}_1, \bar{y}_2). \end{aligned}$$

Thus, we conclude that the images are distinct.

In other case, if $\phi_l(C_i) = \{1\}$ for all $i \in \{1, \dots, 2^l \mid i \text{ even}\}$, then we proceed as follows.

If the function $\phi_{l-1}(C_i) \neq \phi'_2(0, \bar{y}_1, y_{s-l+1}, y_{s-l+2})$, where $(\bar{y}_1, \bar{y}_2) \in C_i$ and $\bar{y}_2 = (y_{s-l+1}, y_{s-l+2}, \dots, y_s)$ for some i . Then, we can write:

$$\begin{aligned} f(0, \dots, 0, 1, 0, \bar{y}_1, \bar{y}_2) &= (0, \dots, 0, 1, 0) \cdot \phi(\bar{y}_1, \bar{y}_2) \oplus g_e(\bar{y}_1, \bar{y}_2) \\ &= \phi_{l-1}(\bar{y}_1, \bar{y}_2) \oplus g_e(\bar{y}_1, \bar{y}_2). \end{aligned}$$

On the other hand, we can express:

$$\begin{aligned} g_l(0, \dots, 0, 1, 0, \bar{y}_1, \bar{y}_2) &= (0, \dots, 0, 1, 0) \cdot \phi'(0, \dots, 0, 1, 0, \bar{y}_1, \bar{y}_2) \oplus g_e(\bar{y}_1, \bar{y}_2) \\ &= \phi'_2(0, \bar{y}_1, y_{s-l+1}, y_{s-l+2}) + g_e(\bar{y}_1, \bar{y}_2). \end{aligned}$$

Therefore, we see that the two images are different.

If $\phi_{l-1}(C_i) = \phi'_2(0, \bar{y}_1, y_{s-l+1}, y_{s-l+2})$, where $(\bar{y}_1, \bar{y}_2) \in C_i$ and $\bar{y}_2 = (y_{s-l+1}, y_{s-l+2}, \dots, y_s)$ for all i . We know that there are 2^{l-2} elements $i \in \{1, \dots, 2^l \mid i \text{ even}\}$ for which $\phi'_2(0, \bar{y}_1, y_{s-l+1}, y_{s-l+2}) = 0$. Then,

$$\begin{aligned} f(0, \dots, 0, 1, 1, \bar{y}_1, \bar{y}_2) &= (0, \dots, 0, 1, 1) \cdot \phi(\bar{y}_1, \bar{y}_2) \oplus g_e(\bar{y}_1, \bar{y}_2) \\ &= \phi_{l-1}(\bar{y}_1, \bar{y}_2) \oplus \phi_l(\bar{y}_1, \bar{y}_2) \oplus g_e(\bar{y}_1, \bar{y}_2) = 0 \oplus 1 \oplus g_e(\bar{y}_1, \bar{y}_2) = 1 \oplus g_e(\bar{y}_1, \bar{y}_2). \end{aligned}$$

On the other and,

$$\begin{aligned} g_l(0, \dots, 0, 1, 1, \bar{y}_1, \bar{y}_2) &= (0, \dots, 0, 1, 1) \cdot \phi'(0, \dots, 0, 1, 1, \bar{y}_1, \bar{y}_2) \oplus g_e(\bar{y}_1, \bar{y}_2) \\ &= \phi'_2(1, \bar{y}_1, y_{s-l+1}, y_{s-l+2}) \oplus \phi'_1(\bar{y}_1, y_{s-l+1}) \oplus g_e(\bar{y}_1, \bar{y}_2) = 1 \oplus 1 \oplus g_e(\bar{y}_1, \bar{y}_2) \\ &= g_e(\bar{y}_1, \bar{y}_2). \end{aligned}$$

Once again, we obtain different images.

Based on the conclusions of all the cases, we find that f and g_l are distinct functions. Therefore, we achieve the desired result.

In the following, we present two algorithms: one for the case where $r = 2$ and the other for twice the case where $r = 1$. These algorithms are primarily derived from analysing the images of the functions ϕ and ϕ' . As we demonstrate, the bent functions generated by each algorithm are distinct in this particular scenario.

We also provide an example for the general case where r can take on different values. By focusing on the specific case of $r = 2$, we derive bent functions over $\mathbb{F}_2^{s+3}$ based on a bent function defined over $\mathbb{F}_2^{s-1}$. We define the mappings of the function ϕ as follows: $\phi(\mathcal{C}_1) = (0, 0)$, $\phi(\mathcal{C}_2) = (1, 1)$, $\phi(\mathcal{C}_3) = (1, 0)$, and $\phi(\mathcal{C}_4) = (0, 1)$. It is important to note that these mappings represent a specific case of the ϕ function.

In Algorithm 1, the union of $\mathcal{A}_3$ and $\mathcal{B}_3$ is the set of all vectors in $\mathbb{F}_2^{s-1}$ with even and odd Hamming weights, respectively, and we are using

$$\mathcal{C}_1 := \begin{matrix} \mathcal{A}_3 & \bar{0} & \bar{0} \\ \mathcal{B}_3 & \bar{1} & \bar{0} \end{matrix}.$$

Algorithm 1 Maiorana-McFarland secondary construction $r = 2$

Input: $s - 1 \geq 2$ even, $g_{s-1}(\bar{y})$, $g_{s-1} : \mathbb{F}_2^{s-1} \rightarrow \mathbb{F}_2$ be a bent function, $\{(x_1, x_2, \bar{y}, y_1, y_2) \in \mathbb{F}_2^{s-1+4} \mid x_1, x_2, y_1, y_2 \in \mathbb{F}_2, \bar{y} \in \mathbb{F}_2^{s-1}\}$

Output: $f(x_1, x_2, \bar{y}, y_1, y_2)$ a bent function, $f|_{\mathcal{A}_1}$ balanced

```

1: for  $x_1, x_2$  from 0 to 1 do
2:   if  $(\bar{y}, y_1, y_2) \in \mathcal{C}_2 := \begin{matrix} \mathcal{A}_3 & \bar{1} & \bar{1} \\ \mathcal{B}_3 & \bar{0} & \bar{1} \end{matrix}$ ,
3:      $[(x_1, x_2) = (0, 1) \text{ and } (x_1, x_2) = (1, 0)]$ 
4:     or
5:      $(\bar{y}, y_1, y_2) \in \mathcal{C}_3 := \begin{matrix} \mathcal{A}_3 & \bar{0} & \bar{1} \\ \mathcal{B}_3 & \bar{1} & \bar{1} \end{matrix}$ ,
6:      $[(x_1, x_2) = (1, 0) \text{ and } (x_1, x_2) = (1, 1)]$ 
7:     or
8:      $(\bar{y}, y_1, y_2) \in \mathcal{C}_4 := \begin{matrix} \mathcal{A}_3 & \bar{1} & \bar{0} \\ \mathcal{B}_3 & \bar{0} & \bar{0} \end{matrix}$ ,
9:      $[(x_1, x_2) = (0, 1) \text{ and } (x_1, x_2) = (1, 1)]$  then
10:     $f(x_1, x_2, \bar{y}, y_1, y_2) = 1 \oplus g_{s-1}(\bar{y})$ ;
11:  else
12:     $f(x_1, x_2, \bar{y}, y_1, y_2) = g_{s-1}(\bar{y})$ ;
13:  end if
14: end for
```

Also, we give an algorithm using $r = 1$ twice (Algorithm 2). Comparing it with Algorithm 1, we can see that we obtain a different bent function.

6 Conclusions

The unique arrangement of the elements in $\mathbb{F}_2^s$ provided us with a clear framework for defining affine spaces. With this advantage, we can demonstrate that

Algorithm 2 Maiorana-McFarland secondary construction two times
 $r = 1$

Input: $s - 1 \geq 2$ even, $g_{s-1}(\bar{y})$, $g_{s-1} : \mathbb{F}_2^{s-1} \rightarrow \mathbb{F}_2$ be a bent function, $\{(x_1, x_2, \bar{y}, y_1, y_2) \in \mathbb{F}_2^{s+3} \mid x_1, x_2, y_1, y_2 \in \mathbb{F}_2, \bar{y} \in \mathbb{F}_2^{s-1}\}$

Output: $g_{s+3}(x_1, x_2, \bar{y}, y_1, y_2)$ a bent function, $g_{s+3}|_{\mathcal{A}_1}$ balanced

```

1: for  $x_1, x_2, y_1, y_2$  from 0 to 1 do
2:   if  $\bar{y}$  is even,
3:      $[(x_1, x_2) = (0, 1) \text{ and } [(y_1, y_2) = (1, 1) \text{ or } (1, 0)]]$  or
4:      $[(x_1, x_2) = (1, 0) \text{ and } [(y_1, y_2) = (0, 1) \text{ or } (1, 0)]]$  or
5:      $[(x_1, x_2) = (1, 1) \text{ and } [(y_1, y_2) = (0, 0) \text{ or } (1, 0)]]$ 
6:     or
7:      $\bar{y}$  is odd,
8:      $[(x_1, x_2) = (0, 1) \text{ and } [(y_1, y_2) = (0, 1) \text{ or } (0, 0)]]$  or
9:      $[(x_1, x_2) = (1, 0) \text{ and } [(y_1, y_2) = (1, 1) \text{ or } (0, 0)]]$  or
10:     $[(x_1, x_2) = (1, 1) \text{ and } [(y_1, y_2) = (1, 0) \text{ or } (0, 0)]]$  then
11:       $g_{s+3}(x_1, x_2, \bar{y}, y_1, y_2) = 1 \oplus g_{s-1}(\bar{y})$ ;
12:   else
13:      $g_{s+3}(x_1, x_2, \bar{y}, y_1, y_2) = g_{s-1}(\bar{y})$ ;
14:   end if
15: end for

```

the affine spaces C_i for $i = 1, \dots, 2^r$ form a partition of $\mathbb{F}_2^s$, along with the other results presented.

The result stated in Theorem 5, that the function on each C_i is a bent function, has not been proven, to our knowledge. This result generalises findings from [2], leading us to Theorem 6, which belongs to the Maiorana-McFarland secondary construction, which is a general case of the construction referenced in [9]. Furthermore, if we aim to construct a bent function with balancedness restricted to the set of vectors with even Hamming weight in its domain, a more thorough analysis along with additional properties is necessary, particularly in contrast to the case when $r = 1$ as discussed in [9].

In Theorem 8, we derive new expressions for bent functions by applying the parameter $r = 1$ a total of l times. Additionally, we present a formula for these functions that maintains balancedness, specifically restricted to the set of vectors with even Hamming weight. The functions obtained through this iteration differ from those derived in Theorem 6, despite both constructions belonging to the class MM_r (which is not limited to bent functions).

The two given algorithms can obtain distinct bent functions incremented in four dimensions relative to the original bent function. We can follow the same procedure to get new bent functions by increasing the dimension by multiples of four.

A desired outcome is to find total balanced functions with high non-linearity, starting with bent functions [8], [12]. In future research, we will explore bent functions derived from the Maiorana-McFarland secondary construction, which demonstrate specifically partial balancedness in our case. By minimising non-linearity as little as possible, we aim to achieve total balancedness.

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